

compute_spatial_mev_basis.R

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compute_spatial_mev_basis
<i>Compute a Reusable Spatial MEM Basis</i>

Description

Computes exact or low-rank Moran eigenvectors and retains explicit state required to project the same basis to held-out locations.

Usage

```
compute_spatial_mev_basis(  
  data_coords_projected = NULL,  
  n_mev = 20L,  
  strategy = "exact",  
  exact_max_locations = 1999L,  
  fast_eigenvectors = 200L,  
  fast_seed = 900723L,  
  exact_function = sjSDM::generateSpatialEV,  
  fast_function = NULL  
)
```

Arguments

data_coords_projected	Data frame with ‘coord_x_km’ and ‘coord_y_km’ columns and unique row names.
n_mev	Positive integer giving the number of public eigenvectors requested.
strategy, exact_max_locations, fast_eigenvectors	Shared strategy settings validated by [resolve_spatial_mev_strategy()].
fast_seed	Positive integer used locally for deterministic NystrC6m knot selection.
exact_function	Exact Moran-eigenvector function. Defaults to [sjSDM::generateSpatialEV()].
fast_function	Fast Moran-eigenvector function. Defaults to [spmoran::meigen_f()].

Details

Exact results retain the current sjSDM values unchanged. Fast results use an exponential-kernel basis and orient each selected eigenvector deterministically. The orientation is retained for matching projection.

Value

Named list with 'data_mev', 'projection_basis', and one-row 'data_provenance'.

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